

CIRCLE DEVELOPER GRANT 2026 · COHORT 2

Plinth

Capital layer for AI trading agents on Arc

Zen Chen · github.com/Ccheh/plinth · MIT, no admin keys

11 contracts live on Arc Testnet · 176/176 tests + 3 stateful invariants · 11-finding pre-deployment audit

The trust gap blocks AI trading from scaling

Agents on Arc run on their own balance. Taking outside capital is blocked by two failures.

1. Custody

Investors must hand keys or signing authority to an unknown agent.

Prior protocols (Enzyme, dHEDGE) custody via multisig — heavy, slow, requires KYC infrastructure.

Plinth: capability-not-custody — agent directs funds, can never withdraw to new addresses.

2. Verification

When an agent reports profit, the only "proof" is the agent's own message on chain.

Off-chain venues (CEXes, Aster, Hyperliquid) leave depositors trusting the agent's word.

Plinth: cryptographic Underwriter reconciles agent's claim against venue's on-chain trade history.

Both must be solved at the infra layer. Plinth is that layer.

Five composable protocols, four compositions live on chain

Each protocol is independent. Each composes with Plinth via on-chain bridges.



Killer feature: cryptographically verifiable PnL

When the trading venue is itself a public chain, "trust my returns" becomes "recomputable from chain."

- 1** **Agent** on Plinth (Arc) creates Vault #5, deploys 0.005 USDC to Aster venue address.
tx 0x7b4d547a... · vault BTC perp momentum strategy
- 2** **Agent** places 3 BTC perp round-trips on Aster L1 mainnet (chainId 1666).
Aster userTrades API · 6 fills total · \$0.13 experimental cost
- 3** **Agent** reports realized PnL on Arc: -0.047207 USDC.
tx 0x0f15c4e1...
- 4** **Independent Underwriter** pulls Aster trades, sums gross +0.0172 USDT – 0.0644 fees = -0.0472 USDT.
verifier reads venue chain directly
- 5** **Matched at 0.00% delta** — Underwriter posts VERIFIED review on chain.
tx 0x7ee06f9c...

Same architecture generalizes to any Arc-native perp DEX (Synthra, Tower Exchange, ...) when their event schemas mature. Grant M5 funds these adapters.

Plinth today + grant funded roadmap

Already shipped, May 2026:

CONTRACTS LIVE 11 on Arc Testnet PlinthV05/V06, 3 venue adapters, 4 bridges (Mandate/Cadence/Crucible/Helm), SponsorPool, MockVenue	TESTS 176/176 + 3 invariants 60K+ random call sequences fuzz-verified · 11-finding pre-deployment audit
ON-CHAIN COMPOSITIONS 4 LIVE Mandate × Plinth · Cadence × Plinth · Crucible × Plinth · Helm × Plinth	PUBLIC-GOODS EXTRACTION 2 OSS deliverables @plinth/verifier-core (npm-ready package) · PlinthSponsorPool (Underwriter sustainability)

Grant funded — \$50K USDC across 5 milestones:

	MILESTONE	AMOUNT
M1	External audit (Trail of Bits) + 5 third-party agent vaults on testnet	\$10K
M2	Production USYC integration on Base via CCTP — real T-bill yield	\$10K
M3	Plinth × Circle Gateway — unified depositor balance across chains	\$10K